JanGyan — Learn Without Limits

Category 3: Road Safety & Public Safety

Module 9 – Modern Road Safety & Smart Mobility

📋 Module Overview

This handbook is your complete, beginner-to-advanced guide to **Modern Road Safety & Smart Mobility** — the technology reshaping how Indians travel, pay tolls, navigate, and stay safe on the road. Whether you drive a petrol car, ride an electric scooter, use a ride-sharing app, or simply want to understand where road safety is heading next, this handbook is written so that you can read it once and feel confident with today's — and tomorrow's — road technology.

By the end of this module, you will not need to search anywhere else to understand modern road safety and smart mobility in India.

🎯 Learning Objectives

After completing this module, you will be able to:

- Understand how smart traffic systems manage congestion and enforcement
- Use FASTag correctly and avoid common toll-related penalties
- Use GPS navigation safely without becoming a distracted driver
- Understand the legal status and value of dashcams in India
- Know the basics of EV safety, including battery standards
- Use ride-sharing services safely, as a passenger or a driver
- Understand how DigiLocker and mParivahan work for digital documents
- Recognise useful smart mobility apps and their safety features
- Understand where road safety technology is headed next
- Understand the basics of how AI is used in traffic management today

🚦 Why This Topic Matters

Road safety in India is no longer only about helmets, seatbelts, and speed limits — it increasingly runs through screens, sensors, and software. FASTag decides how you pay tolls, GPS decides your route, AI-powered cameras decide whether you get a challan, and battery standards decide whether an EV is safe to charge overnight.

> **Reality Check:** Over **95% of national highway toll transactions** in India are now collected electronically through FASTag, and cities like Bengaluru run AI-powered adaptive signals across more than **160 intersections**. At the same time, a wave of electric scooter fires in 2022 forced the government to introduce strict new battery safety standards within months — a reminder that fast-moving technology needs equally fast-moving safety awareness.

This is why understanding modern road technology is now as fundamental to road safety as knowing the traffic signal colours.

📖 Introduction

Modern Road Safety & Smart Mobility covers the technologies that are changing how Indians travel safely — from electronic tolling and AI-driven traffic signals to electric vehicle safety standards and digital documentation. These tools generally fall into two broad groups:

Category	Examples
Infrastructure & Enforcement Technology	Smart traffic systems, FASTag, AI-based ANPR cameras
Personal & Vehicle Technology	GPS navigation, dashcams, EV batteries, ride-sharing apps, digital documents

Used well, this technology makes roads safer and journeys smoother. Used carelessly, it can introduce brand-new risks — a phone mounted in the wrong place, an uncertified EV charger, or blind trust in a navigation app on an unfamiliar road.

🧠 Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the rest of this module is built upon.

Technology Multiplies Habits, It Doesn't Replace Them

> Smart technology amplifies the habits a road user already has. A disciplined driver with GPS navigation reaches their destination more efficiently; a distracted driver with GPS navigation simply finds a new way to be distracted. The technology itself is neutral — the outcome depends on how it is used.

Key Definitions

- **ITMS** — Intelligent Traffic Management System, combining cameras, sensors, and software to monitor and manage traffic
- **ATCS** — Adaptive Traffic Control System, which adjusts signal timing in real time based on actual traffic

- **RFID** — Radio Frequency Identification, the technology behind FASTag's automatic toll payments
- **BMS** — Battery Management System, the electronics that monitor and protect an EV battery pack
- **ANPR** — Automatic Number Plate Recognition, camera-based vehicle identification used in enforcement

🖨 Complete Detailed Explanation

Part A — Smart Traffic Systems

Under the Smart Cities Mission, several Indian cities have deployed **Intelligent Traffic Management Systems (ITMS)** and **Adaptive Traffic Control Systems (ATCS)** that use cameras, sensors, and AI to manage congestion and detect violations automatically. Bengaluru's AI-powered adaptive system, developed with the Centre for Development of Advanced Computing (C-DAC), dynamically adjusts signal timing across more than 160 intersections. These systems typically combine ANPR cameras, red-light and speed violation detection, and a central command centre that can even prioritise green corridors for emergency vehicles.

Part B — FASTag

FASTag is a prepaid RFID tag fixed to a vehicle's windshield, enabling automatic toll deduction without stopping. It has been mandatory for Class M and N vehicles (broadly, cars and larger) on national highways since **15 February 2021**, under the National Highway Fee (Determination of Rates and Collection) Rules, 2008. A vehicle without a valid, working FASTag pays **double the standard toll** in a FASTag lane. NHAI guidelines also promise that if a vehicle waits longer than 10 seconds at a toll booth, the toll is waived, and a yellow line 100 metres before each booth helps manage queuing. A **"One Vehicle, One FASTag"** rule prevents a single tag from being reused across multiple vehicles, and India has begun piloting **GNSS-based satellite tolling**, which could eventually allow barrier-free, distance-based toll collection using GPS instead of physical booths.

Part C — GPS Navigation Safety

GPS navigation has made unfamiliar routes far easier to travel, but it introduces its own risk if used carelessly. Mount your phone or device securely at eye level, set your destination and route before starting to drive, and rely on voice guidance rather than glancing at the screen while moving. Treat any conflict between a navigation app's instruction and an actual road sign or barrier in favour of what you can physically see — apps do not always reflect real-time closures, one-way changes, or construction.

Part D — Dashcams

As of now, no specific Indian law prohibits installing or using a dashcam in a private or commercial vehicle. Dashcam footage can serve as valuable evidence for accident reconstruction, insurance claims, and disputes over traffic stops. Under **Section 63 of the Bharatiya Sakshya Adhiniyam, 2023** (which replaced the Indian Evidence Act), such footage is admissible as electronic evidence provided it was recorded during regular, ordinary use, the device was functioning properly, and it is accompanied by a certificate confirming these conditions and describing the device and process used.

Part E — EV Safety Basics

Following a series of electric two-wheeler fire incidents in 2022, the government tightened India's EV battery safety framework considerably. **AIS-156** governs battery safety for L-category vehicles (electric two- and three-wheelers, quadricycles), while **AIS-038 (Revision 2)** covers M and N category vehicles (cars and larger), both mandatory since December 2022 and strengthened further in 2024. These standards require IP67-rated protection against water and dust, thermal propagation testing to contain a single cell's failure, a microprocessor-based Battery Management System, vibration testing to withstand India's road conditions, and audio-visual warnings for early detection of a thermal event. A related standard, **AIS-138/IS 17017**, governs the safety of EV charging infrastructure itself.

- Charge EVs using only the manufacturer-approved charger, never a generic or modified one
- Avoid charging immediately after a long, hot ride — let the battery cool first
- Do not charge in direct sunlight or in a poorly ventilated space overnight
- Watch for unusual heat, swelling, or smell from the battery pack, and stop use immediately if noticed

Part F — Ride-Sharing Safety

As covered in Module 6, many state transport departments mandate a GPS-linked panic button in app-based cabs and autos, connecting directly to a control room during an emergency. State aggregator rules increasingly require driver verification and clearance of pending traffic challans before a driver can operate on a platform. As a passenger, verify the vehicle and driver details displayed in the app match the vehicle you are boarding, and share your trip status with a trusted contact for late-night journeys.

Part G — Digital Driving Documents (DigiLocker & mParivahan)

As established in Modules 5 and 8, documents such as your driving licence, RC, insurance certificate, and PUC certificate stored in **DigiLocker** or the **mParivahan** app are legally equivalent to physical originals, confirmed by a Press Information Bureau clarification under the Information Technology Act, 2000. Traffic authorities are required to accept these digital versions during routine checks, reducing the risk of penalties for a forgotten physical document.

Part H — Smart Mobility Apps

Beyond navigation and ride-sharing, a growing ecosystem of apps supports safer, smarter travel in India — from real-time public transport

tracking and parking-finder tools to apps that display live traffic incident data drawn from government and crowd-sourced sources. Choosing apps that pull from official or well-established data sources, and keeping them updated, ensures the safety information you rely on is current.

Part I — Future of Road Safety

India's road safety trajectory increasingly points toward systemic, technology-supported prevention rather than reactive enforcement alone — echoing the **Vision Zero** philosophy introduced in Module 1. The **New Delhi Road Safety Declaration** (February 2026) set a long-term national goal of zero preventable road fatalities, aligned with the UN Decade of Action for Road Safety 2021–2030. Emerging areas include connected-vehicle technology (vehicles communicating hazard data to each other), expanded AI-based enforcement, and continued research into advanced driver-assistance features becoming standard even in mass-market vehicles.

Part J — AI in Traffic Management (Overview)

Artificial intelligence already plays a practical, everyday role in Indian traffic management: ANPR cameras identify vehicles for e-challans and stolen-vehicle alerts, computer vision systems detect helmet and seatbelt non-compliance, adaptive signal systems adjust timing based on real-time vehicle density, and centralised command centres use AI-assisted analytics to spot incidents faster and reroute traffic. This is enforcement and planning support, not replacement of human judgement — final decisions on prosecution and infrastructure still rest with traffic police and transport authorities.

📝 Important Terminologies

Term	Simple Meaning
ITMS	Intelligent Traffic Management System, combining sensors and software for traffic control
ATCS	Adaptive Traffic Control System, adjusting signals in real time
FASTag	An RFID tag enabling automatic, stop-free toll payment
GNSS Tolling	Satellite-based tolling that could remove the need for physical toll booths
BMS	Battery Management System, the electronics protecting an EV battery pack
Thermal Runaway	An uncontrolled battery overheating event that can lead to fire
ANPR	Automatic Number Plate Recognition, used for enforcement and identification
ICCC	Integrated Command and Control Centre, a city's central traffic monitoring hub
DigiLocker	A government platform for legally valid digital document storage
Green Corridor	A route where signals are coordinated to give continuous priority, often for emergency vehicles

🧭 Step-by-Step Understanding

Setting Up FASTag Correctly

1. Purchase a FASTag from an authorised bank, NHAI point of sale, or online platform
2. Link it to your specific vehicle's registration number, chassis, and engine details
3. Fix it correctly on the windshield, on the inside, as instructed — not held loosely
4. Keep the linked wallet or bank account funded, and your KYC updated, to avoid blacklisting

Using GPS Navigation Safely

1. Set your destination and confirm the route before you start driving
2. Mount the device securely at eye level, not resting loosely on the dashboard
3. Rely on voice prompts; only glance briefly at the screen when safely stationary
4. Trust visible road signs and closures over the app if the two genuinely conflict

Charging an EV Safely

1. Use only the manufacturer-approved charger and cable
2. Let the battery cool after a long ride before charging
3. Charge in a ventilated, covered space, away from direct sunlight and flammable material
4. Unplug once fully charged rather than leaving it connected indefinitely

🇮🇳 Real-Life Indian Examples

- **Bengaluru's Adaptive Signal Network:** Developed with C-DAC, Bengaluru's AI-powered adaptive traffic control system now covers more than 160 intersections, dynamically adjusting signal timing to reduce delays in one of India's most congested cities.
- **The 2022 EV Fire Incidents:** A string of electric scooter fires across India in the summer of 2022 shook public confidence in electric mobility and directly led to the tightened AIS-156 and AIS-038 battery safety standards introduced within months.
- **FASTag's Rapid National Rollout:** From a mandate in February 2021 to over 8 crore users and roughly 98% penetration within a few years, FASTag stands as one of India's fastest and most successful large-scale road technology rollouts.
- **The New Delhi Road Safety Declaration (2026):** Hosted at IIT Delhi, India's first National Road Safety Confluence set a formal national goal of zero preventable road deaths, aligning domestic policy with the global Decade of Action framework.

💬 Practical Scenarios

Scenario 1: Your FASTag balance runs out just before a toll plaza.
Expect to be charged double the toll in cash — recharge as soon as possible afterward to avoid repeating the issue.

Scenario 2: Your GPS app instructs a U-turn, but you can see a "No U-Turn" sign at that exact spot.
Follow the physical sign, not the app — navigation software doesn't always reflect the latest road rules.

Scenario 3: You want to use dashcam footage to dispute a traffic stop.
Ensure the footage is from a device that was working properly and used regularly, and be prepared to provide the certificate required under Section 63 of the Bharatiya Sakshya Adhiniyam for it to be treated as valid evidence.

Scenario 4: You notice your EV's battery feels unusually warm after fast charging.
Stop charging immediately, disconnect safely if possible, and have it inspected before further use.

Scenario 5: A traffic officer asks for your driving licence, but you only have it saved on DigiLocker.
Show the DigiLocker app version confidently — it is legally equivalent to a physical original and must be accepted.

🏛 Government Rules / Guidelines

- **Smart Cities Mission:** The central initiative under which many Indian cities have deployed ITMS and ATCS traffic technology.
- **National Highway Fee (Determination of Rates and Collection) Rules, 2008:** The legal basis for FASTag's mandatory use and the double-toll penalty for non-compliance.
- **AIS-156 & AIS-038 (Rev. 2):** Mandatory EV battery safety standards administered through ARAI-accredited testing.
- **Bharatiya Sakshya Adhiniyam, 2023:** Governs the admissibility of dashcam and other electronic evidence in Indian courts.
- **DigiLocker / mParivahan:** Government platforms providing legally recognised digital storage for driving and vehicle documents.
- **New Delhi Road Safety Declaration (2026):** A national commitment to zero preventable road fatalities, aligned with the UN Decade of Action.

⚖ Laws & Policies

KEY LEGAL & REGULATORY PROVISIONS

Provision	What It Covers
NH Fee Rules, 2008	Legal basis for mandatory FASTag and double-toll penalties
AIS-156	Battery safety standard for electric two/three-wheelers and quadricycles
AIS-038 (Rev. 2)	Battery safety standard for electric cars and larger vehicles
AIS-138 / IS 17017	Safety standard for EV charging infrastructure
Section 63, Bharatiya Sakshya Adhiniyam, 2023	Admissibility of dashcam and other electronic evidence in court

There is currently no dedicated national law specifically regulating dashcams, EV charging behaviour at home, or GPS use while driving beyond the existing general provisions against dangerous or distracted driving under Section 184 of the Motor Vehicles Act — good practice in these newer areas is often guided by manufacturer instructions and common sense as much as by statute.

📊 Important Statistics (India)

Indicator	Figure
National highway toll collected electronically via FASTag	Over 95% of transactions
FASTag users nationally	Over 8 crore, roughly 98% penetration
FASTag mandatory since	15 February 2021
Bengaluru's AI-adaptive signal coverage	160+ intersections
EV battery safety standards mandatory since	December 2022 (strengthened further, 2024)
Maximum toll-booth wait before fee is waived	10 seconds (NHAI guideline)

✓ Best Practices

- Keep your FASTag funded and KYC updated to avoid blacklisting and double charges
- Set your GPS route before driving, not while already moving
- Use a properly mounted dashcam and preserve footage promptly after any incident
- Charge EVs only with manufacturer-approved equipment, in a ventilated space
- Verify ride-sharing vehicle and driver details before boarding
- Keep digital copies of all key documents on DigiLocker or mParivahan as backup
- Stay informed about evolving road safety technology and standards

✕ Common Mistakes

- Letting a FASTag balance run out unnoticed before a long trip
- Interacting with a GPS app's screen manually while driving
- Assuming dashcam footage is automatically valid evidence without proper certification
- Using a non-standard or damaged charger for an EV
- Ignoring unusual heat or smell from an EV battery pack
- Boarding a ride-share vehicle without checking it matches the app details
- Relying solely on a navigation app's instruction over a visible road sign

🔍 Myths vs Facts

Myth	Fact
"Dashcams are illegal or restricted in India"	No specific law prohibits their use in private or commercial vehicles
"All EV battery fires are simply manufacturing defects"	Many incidents also trace back to improper charging practices or non-standard chargers
"FASTag balance issues are rare and don't matter much"	Insufficient balance or expired KYC leads directly to double toll charges
"AI traffic cameras replace the need for traffic police"	AI systems support detection and monitoring, but enforcement decisions remain with human authorities
"A GPS app always knows the current road situation"	Apps can lag behind real-time closures, diversions, or new road signage

📌 Do's and Don'ts

Do	Don't
Keep your FASTag funded and KYC current	Don't assume a fitted FASTag always has sufficient balance
Set navigation routes before starting your journey	Don't type or scroll on a GPS app while driving
Use manufacturer-approved EV chargers only	Don't use generic, damaged, or modified chargers
Verify ride-share vehicle details before boarding	Don't board a vehicle that doesn't match the app's details
Keep documents backed up on DigiLocker/mParivahan	Don't rely solely on a single physical copy
Follow visible road signs over app instructions when they conflict	Don't blindly follow GPS directions into a clearly closed or restricted road

🛡 Safety Tips

Tolls & Navigation

- Recharge your FASTag before a long trip, not en route
- Pre-plan alternate routes for areas with known poor network coverage

EVs & Charging

- Avoid fully draining an EV battery repeatedly; charge before it gets critically low
- Keep the charging area clear of flammable materials

Digital Tools

- Keep your dashcam's date and time settings accurate for reliable evidence
- Update mobility and navigation apps regularly for the latest map and safety data

📋 Practical Checklist

- [ ] My FASTag is funded, correctly fitted, and KYC-updated
- [ ] I set my GPS route before driving, not while in motion
- [ ] My dashcam, if fitted, is working and its footage is regularly backed up
- [ ] I charge my EV only with an approved charger, in a safe location
- [ ] I verify ride-share vehicle and driver details before boarding
- [ ] My key documents are backed up on DigiLocker or mParivahan
- [ ] I follow visible road signs over app instructions when they conflict
- [ ] I stay reasonably informed about new road safety technology and rules

🎓 Quick Revision Notes

- FASTag mandatory since 15 February 2021; non-functional or missing tag means double toll
- Smart traffic systems (ITMS/ATCS) use AI, sensors, and cameras to manage congestion and enforcement
- Dashcams are legal in India; footage needs a Section 63 (BSA, 2023) certificate to be admissible evidence
- EV battery safety: AIS-156 (two/three-wheelers), AIS-038 Rev. 2 (cars and larger), mandatory since Dec 2022
- DigiLocker/mParivahan documents are legally equivalent to physical originals
- AI in traffic management supports detection and monitoring; humans retain enforcement authority
- New Delhi Road Safety Declaration (2026) set a national goal of zero preventable road deaths

🎯 Key Takeaways

1. Modern road technology is now a core part of everyday road safety in India, not an optional extra
2. FASTag, dashcams, and digital documents each carry specific rules worth knowing well
3. EV safety depends as much on user charging habits as on manufacturing standards
4. AI-powered traffic systems are expanding rapidly, but remain tools that support, not replace, human oversight
5. Technology only improves safety when paired with the same good judgement covered throughout this platform
6. Staying current with evolving standards is now a genuine, ongoing part of responsible road use

📖 Glossary

- **ANPR** — Automatic Number Plate Recognition, camera-based vehicle identification
- **ATCS** — Adaptive Traffic Control System, adjusting signals in real time
- **BMS** — Battery Management System, protecting an EV battery pack
- **FASTag** — An RFID tag enabling automatic, stop-free toll payment
- **GNSS Tolling** — Satellite-based tolling without physical toll booths
- **ICCC** — Integrated Command and Control Centre, a city's traffic monitoring hub
- **ITMS** — Intelligent Traffic Management System
- **RFID** — Radio Frequency Identification technology
- **Thermal Runaway** — Uncontrolled battery overheating that can lead to fire
- **Vision Zero** — A safe-system approach aiming to eliminate road deaths and serious injuries

? Frequently Asked Questions

1. Since when has FASTag been mandatory in India?

Since 15 February 2021, for Class M and N vehicles on national highways.

2. What happens if I don't have a working FASTag at a toll plaza?

You are charged double the standard toll rate.

3. What is the "One Vehicle, One FASTag" rule?

A rule preventing a single FASTag from being reused or linked across multiple vehicles.

4. Are dashcams legal in India?

Yes, there is no specific law prohibiting their use in private or commercial vehicles.

5. Is dashcam footage admissible in an Indian court?

Yes, under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023, provided it meets certain authenticity conditions and is accompanied by a certificate.

6. What standard governs EV battery safety for electric two-wheelers?

AIS-156, mandatory since December 2022.

7. What standard governs battery safety for electric cars?

AIS-038 (Revision 2), also mandatory since December 2022.

8. Why were India's EV battery safety standards tightened?

Following a series of electric two-wheeler fire incidents in 2022 that raised serious safety concerns.

9. Should I use any charger with my EV?

No, only the manufacturer-approved charger and cable should be used.

10. What is an Adaptive Traffic Control System (ATCS)?

A system that adjusts traffic signal timing in real time based on actual traffic conditions, rather than a fixed schedule.

11. Which Indian city has a well-known AI-powered adaptive signal system?

Bengaluru, whose system covers more than 160 intersections.

12. Are DigiLocker and mParivahan documents legally valid?

Yes, they are legally equivalent to physical originals under the Information Technology Act, 2000.

13. What should I check before boarding an app-based cab or auto?

That the vehicle registration number and driver details match what is shown in the app.

14. What is a GPS-linked panic button?

A feature in public service vehicles that sends an SOS signal with location data to a control room when pressed.

15. Is it safe to interact with a GPS app while driving?

No, you should set your route before driving and rely on voice guidance rather than looking at or touching the screen while moving.

16. What is GNSS-based tolling?

A satellite-based tolling approach being piloted in India that could eventually remove the need for physical toll booths.

17. Does AI fully replace traffic police in India?

No, AI systems support detection, monitoring, and enforcement, but final decisions remain with human traffic authorities.

18. What is the New Delhi Road Safety Declaration?

A 2026 national commitment, made at IIT Delhi's Road Safety Confluence, targeting zero preventable road deaths, aligned with the UN Decade of Action.

19. What does IP67 mean for an EV battery?

It indicates the battery pack is protected against dust ingress and can withstand temporary submersion in water.

20. What should I do if my EV battery feels unusually hot?

Stop using or charging it immediately and have it inspected before further use.

21. What is a "green corridor" in traffic management?

A route where signals are coordinated to give continuous priority, often used to speed up emergency vehicle response.

22. Where can I learn more about official Indian EV safety standards?

Through the Automotive Research Association of India (ARAI) and Ministry of Road Transport and Highways publications on Automotive Industry Standards.

🖼 Suggested Infographics, Flowcharts & Illustrations

(For the JanGyan design team — placement suggestions only; no images generated here)

- **Smart traffic system architecture diagram** (cameras, ATCS, ICCC) → within "Smart Traffic Systems"
- **FASTag "how it works" diagram** → within "FASTag"
- **EV battery safety feature breakdown** → within "EV Safety Basics"
- **Flowchart: "Is My Dashcam Footage Admissible?"** → within "Dashcams"
- **Digital documents vs. physical documents comparison card** → within "Digital Driving Documents"
- **Road safety technology timeline** (past → present → future) → within "Future of Road Safety"

🖇 Official References

- National Highways Authority of India (NHAI) — FASTag rules and toll collection guidelines
- Automotive Research Association of India (ARAI) — AIS-156 and AIS-038 (Rev. 2) battery safety standards
- Ministry of Road Transport and Highways (MoRTH) — Automotive Industry Standards and road safety policy
- Bharatiya Sakshya Adhiniyam, 2023 — provisions on electronic evidence admissibility

- Smart Cities Mission, Ministry of Housing and Urban Affairs — ITMS/ATCS deployment guidance
- Press Information Bureau (PIB) — clarification on DigiLocker/mParivahan document validity

🔗 Official Government Website Links

- Ministry of Road Transport & Highways: **https://morth.nic.in**
- National Highways Authority of India: **https://nhai.gov.in**
- Parivahan Sewa: **https://parivahan.gov.in**
- DigiLocker: **https://digilocker.gov.in**
- Smart Cities Mission: **https://smartcities.gov.in**

Emergency Numbers: All-in-one Police/Ambulance/Fire: **112** · NHAI Highway Helpline: **1033**

🏁 Conclusion

India's roads are changing faster than at any point in recent memory — electronic tolls have replaced long queues, AI-powered cameras watch over busy intersections, electric vehicles are becoming everyday transport, and a phone in your pocket now carries your driving licence. None of this technology makes safe road use automatic; it simply gives prepared, informed citizens better tools to work with.

This is only the beginning of your journey with JanGyan into the future of road safety. Stay curious, stay updated, and remember: **the safest road user is not the one with the most technology, but the one who understands it best.**

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